SQL BASICS

Presented by :

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INTRODUCTION

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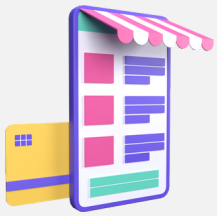


Online Shopping

We are analyzing a small online shopping platform where customers register and place orders. The company wants to track customer details and their purchases for reporting and analysis.

- Database Name: RetailDB_1
- Purpose: To understand who the customers are, when they joined, and what orders they placed.
- At this stage, we are only storing customer information and their orders.

Tables in RetailDB:

Customers – stores details of each registered customer. (**customer_id**, name, email, city, signup_date)

Products - holds all the products details (product_id, **product_name**, category, price)

Orders – keeps records of every order placed, linked to the customer. (**order_id**, customer_id, product_id, order_date, quantity, total_amount, payment_mode)

Orders

order_id	customer_id	product_id	order_date	quantity	total_amount	payment_mode
1	1	1	2024-07-10	1	79999.00	UPI
2	2	2	2024-07-15	1	74999.00	Credit Card
3	3	3	2024-07-20	1	4999.00	Net Banking
4	4	6	2024-07-22	2	5998.00	Debit Card
5	5	5	2024-08-01	1	1999.00	UPI
6	6	7	2024-08-03	1	6499.00	Credit Card
7	8	10	2024-08-05	1	59999.00	Debit Card
8	9	8	2024-08-10	3	1497.00	UPI

Other Data

Customers

customer_id	name	email	city	signup_date
1	Rohit Kumar	rohit.kumar@gmail.com	Delhi	2024-01-15
2	Sneha Sharma	sneha.sharma@yahoo.com	Mumbai	2024-02-20
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2024-03-05
4	Priya Reddy	priya.reddy@gmail.com	Hyderabad	2024-03-22
5	Karan Singh	karan.singh@outlook.com	Chennai	2024-04-10
6	Neha Verma	neha.verma@gmail.com	Pune	2024-05-08
7	Arjun Mehta	arjun.mehta@gmail.com	Jaipur	2024-05-15
8	Ananya Iyer	ananya.iyer@gmail.com	Bengaluru	2024-06-02
9	Vikram Das	vikram.das@gmail.com	Kolkata	2024-06-20
10	Meera Nair	meera.nair@gmail.com	Kochi	2024-07-01

Products

product_id	product_name	category	price
1	iPhone 15	Electronics	79999.00
2	Samsung Galaxy S24	Electronics	74999.00
3	Noise Smartwatch	Wearables	4999.00
4	Boat Earbuds	Wearables	2499.00
5	Kurta Set	Fashion	1999.00
6	Running Shoes	Fashion	2999.00
7	Prestige Mixer Grinder	Home Appliances	6499.00
8	Tata Tea 1kg Pack	Groceries	499.00
9	Amul Butter 500g	Groceries	285.00
10	Sony Bravia 55” TV	Electronics	59999.00

Task: Solve the below mentioned questions by writing SQL queries

1. Fetch all customers from the database.

2. Show only the customer names and their cities.

3. Find customers who live in Mumbai.

4. Get all orders placed after 1st August 2024.

5. List all products priced greater than ₹5000.

6. Count how many customers exist in the system.

7. Update a customer's city (e.g., change Rohit Kumar's city to Hyderabad).

8. Delete an order (e.g., remove order with ID = 5).

9. Display product names with their original price and price increased by 10%.

10. Show only the unique cities where customers live.

11. Get the first 3 customers who signed up.

12. Skip the first 2 customers and fetch the next 3 customers.

13. Find products with prices between ₹2000 and ₹6000.

14. Find customers who are from Mumbai OR Chennai.

15. Find customers who are NOT from Delhi.

16. Find orders that are NOT paid by UPI.

17. Get the average order amount across all orders.

18. Show the highest order amount.

19. Show the lowest product price.

20. Find the total money spent across all orders

Table creation and Import of data from csv file

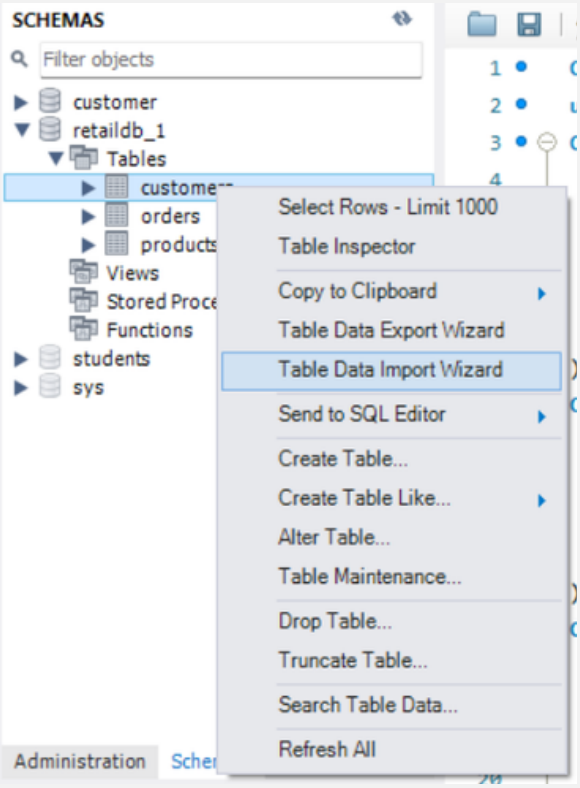
Step 1: Database creation

```
CREATE database RetailDB_1;
use retaildb_1;
```

Step 2: Table creation (Customer)

```
CREATE TABLE Customers (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(150) UNIQUE,
    city VARCHAR(50),
    signup_date DATE
);
```

Step 3: Import data from CSV file



Step 4: Display Imported data

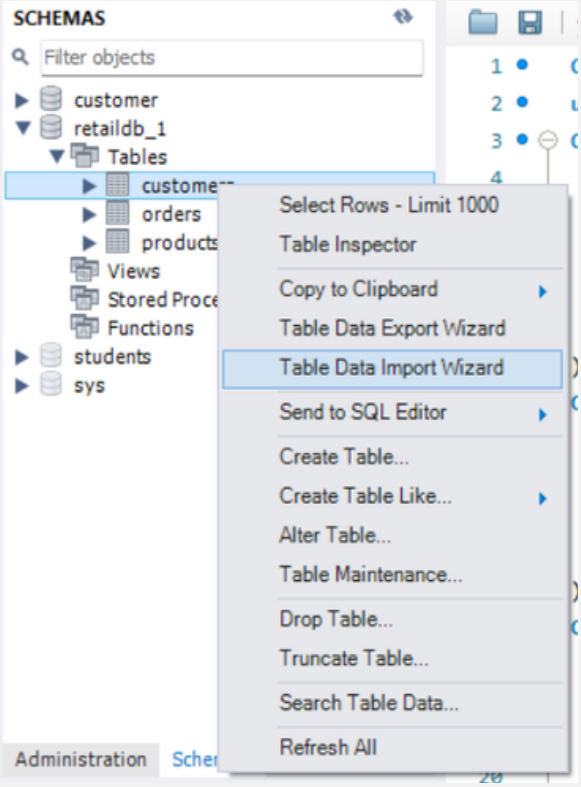
customer_id	name	email	city	signup_date
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2023-02-18
4	Priya Reddy	priya.reddy@gmail.com	Hyderabad	2023-03-02
5	Karan Singh	karan.singh@outlook.com	Chennai	2023-03-15
6	Neha Verma	neha.verma@gmail.com	Pune	2023-04-01
7	Arjun Mehta	arjun.mehta@gmail.com	Bengaluru	2023-04-20
8	Ritika Gupta	ritika.gupta@yahoo.com	Kolkata	2023-05-12
9	Vikram Joshi	vikram.joshi@gmail.com	Lucknow	2023-05-25
10	Ananya Das	ananya.das@gmail.com	Bhubaneswar	2023-06-08
11	Suresh Iyer	suresh.iyer@gmail.com	Chennai	2023-06-20
12	Megha Kapoor	megha.kapoor@yahoo.com	Jaipur	2023-07-03
13	Ravi Shankar	ravi.shankar@gmail.com	Delhi	2023-07-15
14	Tanya Mishra	tanya.mishra@gmail.com	Noida	2023-08-01
15	Aditya Jain	aditya.jain@gmail.com	Indore	2023-08-14

Table creation and Import of data from csv file

Step 5: Table creation (Orders)

```sql
CREATE TABLE Orders (
  order_id INT AUTO_INCREMENT PRIMARY KEY,
  customer_id INT,
  product_id INT,
  order_date DATE,
  quantity INT,
  total_amount DECIMAL(10,2),
  payment_mode VARCHAR(50),
  FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
  FOREIGN KEY (product_id) REFERENCES Products(product_id)
);
```

Step 6: Import data from CSV file



Step 7: Display Imported data

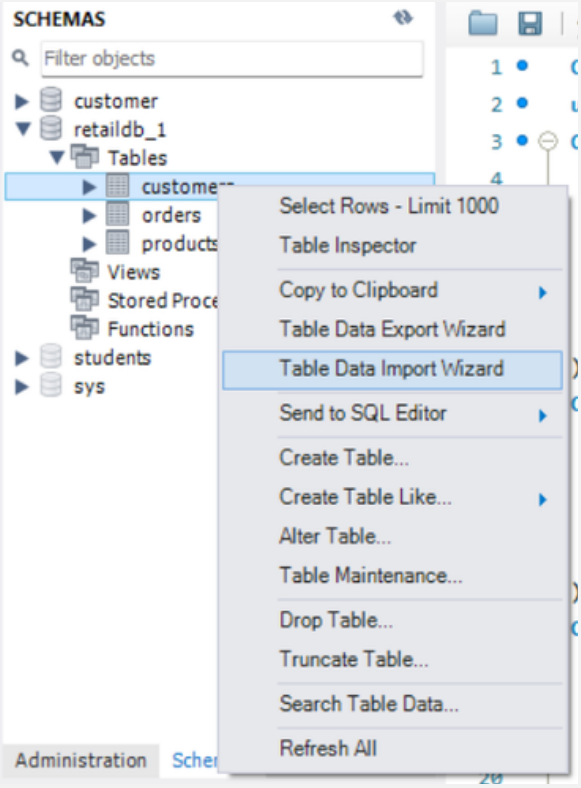
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_id	product_id	order_date	quantity	total_amount	payment_mode
1	1	1	2024-01-05	1	79999	UPI
2	2	2	2024-01-10	1	74999	Credit Card
3	3	3	2024-01-15	2	9998	Debit Card
4	4	4	2024-01-18	1	2499	UPI
5	5	5	2024-01-20	3	5997	Cash
6	6	6	2024-01-22	1	2999	UPI
7	7	7	2024-01-25	1	6499	Credit Card
8	8	8	2024-02-01	1	59999	Net Banking
9	9	9	2024-02-05	1	55999	UPI
10	10	10	2024-02-07	2	2998	Debit Card
11	11	1	2024-02-10	1	79999	Credit Card
12	12	2	2024-02-12	1	74999	Net Banking
13	13	3	2024-02-14	1	4999	UPI
14	14	4	2024-02-16	2	4998	Cash
15	15	5	2024-02-20	1	1999	UPI
16	6	6	2024-02-25	2	5998	Credit Card
17	7	7	2024-03-01	1	6499	UPI
18	8	8	2024-03-05	1	59999	Debit Card
19	9	9	2024-03-07	2	111998	Credit Card
20	10	10	2024-03-10	1	1499	Cash
21	11	5	2024-03-12	2	3998	UPI
22	12	3	2024-03-15	1	4999	Net Banking
23	13	4	2024-03-18	3	7497	Credit Card
24	14	6	2024-03-20	1	2999	UPI
25	15	1	2024-03-25	1	79999	Debit Card
26	2	7	2024-03-28	1	6499	Credit Card

Step 8: Table creation (Products)

```sql
CREATE TABLE Products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    price DECIMAL(10,2)
);
```

Step 9: Import data from CSV file



Step 10: Display Imported data

product_id	product_name	category	price
1	iPhone 15	Electronics	79999.00
2	Samsung Galaxy S24	Electronics	74999.00
3	Noise Smartwatch	Wearables	4999.00
4	Boat Earbuds	Wearables	2499.00
5	Kurta Set	Fashion	1999.00
6	Running Shoes	Fashion	2999.00
7	Prestige Mixer Grinder	Home Appliances	6499.00
8	Sony Bravia 55Ã¢Â€ TV	Electronics	59999.00
9	Lenovo Laptop	Electronics	55999.00
10	Philips Trimmer	Personal Care	1499.00

Solved problem statements

1. Fetch all customers from the database

```
1 •   SELECT * FROM retaildb_1.customers;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

customer_id	name	email	city	signup_date
1	Rohit Kumar	rohit.kumar@gmail.com	Delhi	2023-01-12
2	Sneha Sharma	sneha.sharma@yahoo.com	Mumbai	2023-02-05
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2023-02-18
4	Priya Reddy	priya.reddy@gmail.com	Hyderabad	2023-03-02
5	Karan Singh	karan.singh@outlook.com	Chennai	2023-03-15
6	Neha Verma	neha.verma@gmail.com	Pune	2023-04-01
7	Arjun Mehta	arjun.mehta@gmail.com	Bengaluru	2023-04-20
8	Ritika Gupta	ritika.gupta@yahoo.com	Kolkata	2023-05-12
9	Vikram Joshi	vikram.joshi@gmail.com	Lucknow	2023-05-25
10	Ananya Das	ananya.das@gmail.com	Bhubaneswar	2023-06-08
11	Suresh Iyer	suresh.iyer@gmail.com	Chennai	2023-06-20
12	Megha Kapoor	megha.kapoor@yahoo.com	Jaipur	2023-07-03
13	Ravi Shankar	ravi.shankar@gmail.com	Delhi	2023-07-15
14	Tanya Mishra	tanya.mishra@gmail.com	Noida	2023-08-01
15	Aditya Jain	aditya.jain@gmail.com	Indore	2023-08-14
NULL	NULL	NULL	NULL	NULL

2. Show only the customer names and their cities.

```
1 •   SELECT name, city FROM retaildb_1.customers;
```

Result Grid | Filter Rows: | Export: | Wra

name	city
Rohit Kumar	Delhi
Sneha Sharma	Mumbai
Amit Patel	Ahmedabad
Priya Reddy	Hyderabad
Karan Singh	Chennai
Neha Verma	Pune
Arjun Mehta	Bengaluru
Ritika Gupta	Kolkata
Vikram Joshi	Lucknow
Ananya Das	Bhubaneswar
Suresh Iyer	Chennai
Megha Kapoor	Jaipur
Ravi Shankar	Delhi
Tanya Mishra	Noida
Aditya Jain	Indore

3. Find customers who live in Mumbai.

```
1 •  SELECT name, city FROM retaildb_1.customers where city="Mumbai";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	city
Sneha Sharma	Mumbai

4. Get all orders placed after 1st August 2024.

```
1 •  SELECT * FROM retaildb_1.orders Where order_date > "2024-01-08";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_id	product_id	order_date	quantity	total_amount	payment_mode
4	4	4	2024-01-18	1	2499	UPI
5	5	5	2024-01-20	3	5997	Cash
6	6	6	2024-01-22	1	2999	UPI
7	7	7	2024-01-25	1	6499	Credit Card
8	8	8	2024-02-01	1	59999	Net Banking
9	9	9	2024-02-05	1	55999	UPI
10	10	10	2024-02-07	2	2998	Debit Card
11	11	1	2024-02-10	1	79999	Credit Card
12	12	2	2024-02-12	1	74999	Net Banking
13	13	3	2024-02-14	1	4999	UPI
14	14	4	2024-02-16	2	4998	Cash
15	15	5	2024-02-20	1	1999	UPI
16	6	6	2024-02-25	2	5998	Credit Card
17	7	7	2024-03-01	1	6499	UPI
18	8	8	2024-03-05	1	59999	Debit Card
19	9	9	2024-03-07	2	111998	Credit Card
20	10	10	2024-03-10	1	1499	Cash
21	11	5	2024-03-12	2	3998	UPI
22	12	3	2024-03-15	1	4999	Net Banking
23	13	4	2024-03-18	3	7497	Credit Card

5. List all products priced greater than ₹5000.

```
1 •  SELECT * FROM retaildb_1.products where price > 5000;
```

Result Grid | Filter Rows: | Edit: | Export/

product_id	product_name	category	price
1	iPhone 15	Electronics	79999.00
2	Samsung Galaxy S24	Electronics	74999.00
7	Prestige Mixer Grinder	Home Appliances	6499.00
8	Sony Bravia 55Ã¢Â€ TV	Electronics	59999.00
9	Lenovo Laptop	Electronics	55999.00
NULL	NULL	NULL	NULL

6. Count how many customers exist in the system.

```
SELECT count(customer_id) FROM retaildb_1.customers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

count(customer_id)
15

7. Update a customer's city (e.g., change Rohit Kumar's city to Hyderabad).

```
set sql_safe_updates=0;
update retaildb_1.customers set city='Hyderabad' where name='Rohit Kumar';
select * from retaildb_1.customers where name='Rohit Kumar';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

customer_id	name	email	city	signup_date
1	Rohit Kumar	rohit.kumar@gmail.com	Hyderabad	2023-01-12
NULL	NULL	NULL	NULL	NULL

8. Delete an order (e.g., remove order with ID = 5).

```
delete FROM retaildb_1.orders Where order_id = 5;
select * from retaildb_1.orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_id	product_id	order_date	quantity	total_amount	payment_mode
1	1	1	2024-01-05	1	79999	UPI
2	2	2	2024-01-10	1	74999	Credit Card
3	3	3	2024-01-15	2	9998	Debit Card
4	4	4	2024-01-18	1	2499	UPI
6	6	6	2024-01-22	1	2999	UPI
7	7	7	2024-01-25	1	6499	Credit Card
8	8	8	2024-02-01	1	59999	Net Banking
9	9	9	2024-02-05	1	55999	UPI
10	10	10	2024-02-07	2	2998	Debit Card
11	11	1	2024-02-10	1	79999	Credit Card

9. Display product names with their original price and price increased by 10%.

```
-- 9. Display product names with their original price and price increased by 10%
SELECT *, (price*1.10) as "increased price" FROM retaildb_1.products;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_name	category	price	increased price
1	iPhone 15	Electronics	79999.00	87998.9000
2	Samsung Galaxy S24	Electronics	74999.00	82498.9000
3	Noise Smartwatch	Wearables	4999.00	5498.9000
4	Boat Earbuds	Wearables	2499.00	2748.9000
5	Kurta Set	Fashion	1999.00	2198.9000
6	Running Shoes	Fashion	2999.00	3298.9000
7	Prestige Mixer Grinder	Home Appliances	6499.00	7148.9000
8	Sony Bravia 55Ã¢Â€ TV	Electronics	59999.00	65998.9000
9	Lenovo Laptop	Electronics	55999.00	61598.9000
10	Philips Trimmer	Personal Care	1499.00	1648.9000

10.Show only the unique cities where customers live.

```
select distinct city from retaildb_1.customers;
```

Result Grid | Filter Rows: | Export: | Wrap C

city
Hyderabad
Mumbai
Ahmedabad
Chennai
Pune
Bengaluru
Kolkata
Lucknow
Bhubaneswar
Jaipur
Delhi
Noida
Indore

11. Get the first 3 customers who signed up.

```
select * from retaildb_1.customers limit 3;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

customer_id	name	email	city	signup_date
1	Rohit Kumar	rohit.kumar@gmail.com	Hyderabad	2023-01-12
2	Sneha Sharma	sneha.sharma@yahoo.com	Mumbai	2023-02-05
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2023-02-18
NULL	NULL	NULL	NULL	NULL

12. Skip the first 2 customers and fetch the next 3 customers.

```
select * from retaildb_1.customers where customer_id > 2 limit 3;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

customer_id	name	email	city	signup_date
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2023-02-18
4	Priya Reddy	priya.reddy@gmail.com	Hyderabad	2023-03-02
5	Karan Singh	karan.singh@outlook.com	Chennai	2023-03-15
NULL	NULL	NULL	NULL	NULL

13. Find products with prices between ₹2000 and ₹6000.

```
SELECT * FROM retaildb_1.products where price > 2000 and price < 6000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

product_id	product_name	category	price
3	Noise Smartwatch	Wearables	4999.00
4	Boat Earbuds	Wearables	2499.00
6	Running Shoes	Fashion	2999.00
NULL	NULL	NULL	NULL

14. Find customers who are from Mumbai OR Chennai.

```
select * from retaildb_1.customers where city = 'Mumbai' or city = 'Chennai';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

customer_id	name	email	city	signup_date
2	Sneha Sharma	sneha.sharma@yahoo.com	Mumbai	2023-02-05
5	Karan Singh	karan.singh@outlook.com	Chennai	2023-03-15
11	Suresh Iyer	suresh.iyer@gmail.com	Chennai	2023-06-20
NULL	NULL	NULL	NULL	NULL

15. Find customers who are NOT from Delhi.

```
select * from retaildb_1.customers where city != 'Delhi';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

customer_id	name	email	city	signup_date
1	Rohit Kumar	rohit.kumar@gmail.com	Hyderabad	2023-01-12
2	Sneha Sharma	sneha.sharma@yahoo.com	Mumbai	2023-02-05
3	Amit Patel	amit.patel@gmail.com	Ahmedabad	2023-02-18
4	Priya Reddy	priya.reddy@gmail.com	Hyderabad	2023-03-02
5	Karan Singh	karan.singh@outlook.com	Chennai	2023-03-15
6	Neha Verma	neha.verma@gmail.com	Pune	2023-04-01
7	Arjun Mehta	arjun.mehta@gmail.com	Bengaluru	2023-04-20
8	Ritika Gupta	ritika.gupta@yahoo.com	Kolkata	2023-05-12
9	Vikram Joshi	vikram.joshi@gmail.com	Lucknow	2023-05-25
10	Ananya Das	ananya.das@gmail.com	Bhubaneswar	2023-06-08
11	Suresh Iyer	suresh.iyer@gmail.com	Chennai	2023-06-20
12	Megha Kapoor	megha.kapoor@yahoo.com	Jaipur	2023-07-03
14	Tanya Mishra	tanya.mishra@gmail.com	Noida	2023-08-01
15	Aditya Jain	aditya.jain@gmail.com	Indore	2023-08-14
NULL	NULL	NULL	NULL	NULL

16. Find orders that are NOT paid by UPI.

```
select * from retaildb_1.orders where payment_mode != 'UPI';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_id	product_id	order_date	quantity	total_amount	payment_mode
2	2	2	2024-01-10	1	74999	Credit Card
3	3	3	2024-01-15	2	9998	Debit Card
7	7	7	2024-01-25	1	6499	Credit Card
8	8	8	2024-02-01	1	59999	Net Banking
10	10	10	2024-02-07	2	2998	Debit Card
11	11	1	2024-02-10	1	79999	Credit Card
12	12	2	2024-02-12	1	74999	Net Banking
14	14	4	2024-02-16	2	4998	Cash
16	6	6	2024-02-25	2	5998	Credit Card
18	8	8	2024-03-05	1	59999	Debit Card
19	9	9	2024-03-07	2	111998	Credit Card
20	10	10	2024-03-10	1	1499	Cash
22	12	3	2024-03-15	1	4999	Net Banking
23	13	4	2024-03-18	3	7497	Credit Card
25	15	1	2024-03-25	1	79999	Debit Card
26	2	7	2024-03-28	1	6499	Credit Card
28	9	2	2024-04-03	1	74999	Cash
29	13	9	2024-04-05	1	55999	Net Banking

17. Get the average order amount across all orders.

```
1 •   select avg(total_amount) from retaildb_1.orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	avg(total_amount)
▶	32722.8276

18. Show the highest order amount.

```
1 •   select max(total_amount) from retaildb_1.orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cel

	max(total_amount)
▶	111998

19. Show the lowest product price.

```
1 •   select min(total_amount) from retaildb_1.orders;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

	min(total_amount)
▶	1499

20. Find the total money spent across all orders.

```
1 •   select sum(total_amount) from retaildb_1.orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cel

	sum(total_amount)
▶	948962

Thank you!